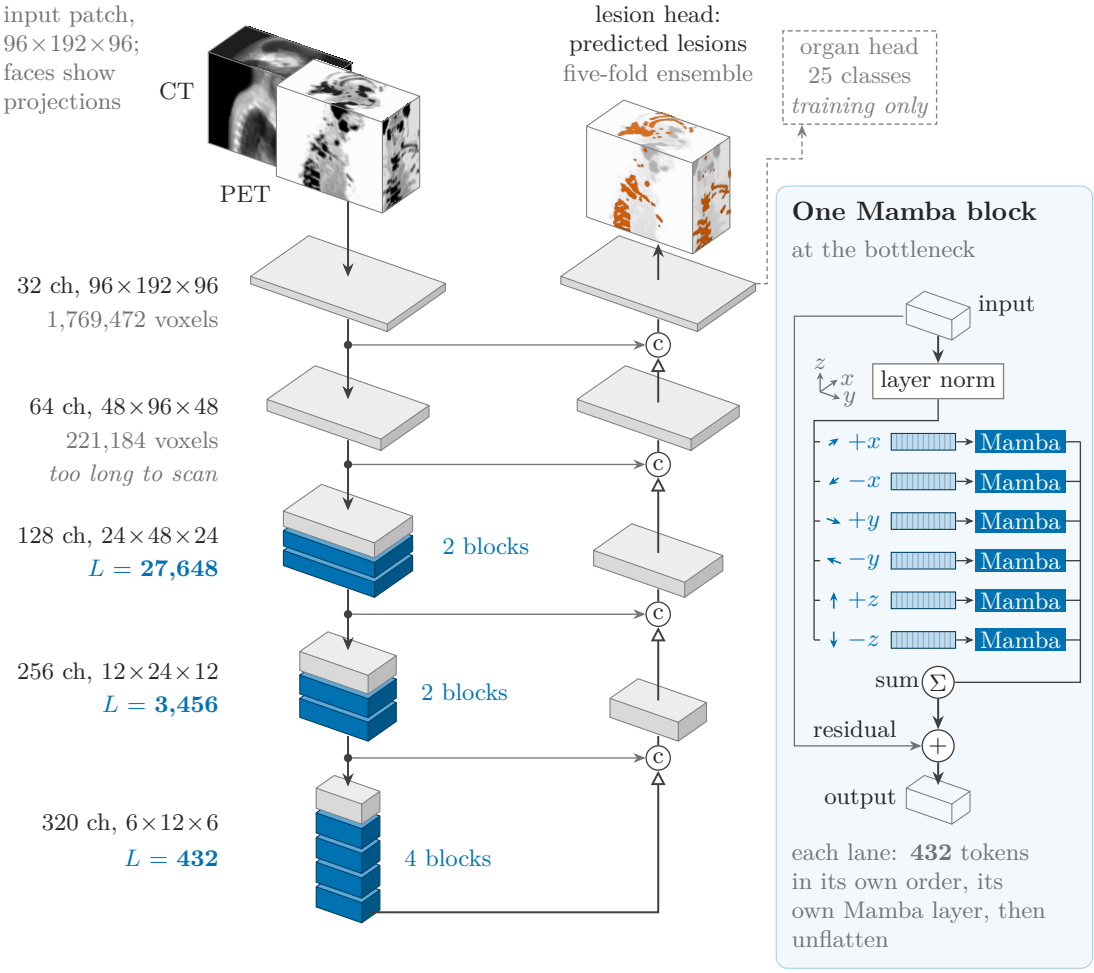

input patch,
96×192×96;
faces show
projections
CT
PET
lesion head:
predicted lesions
five-fold ensemble
organ head
25 classes
training only
32 ch, 96×192×96
1,769,472 voxels
64 ch, 48×96×48
221,184 voxels
too long to scan
128 ch, 24×48×24
L = 27,648
2 blocks
256 ch, 12×24×12
L = 3,456
2 blocks
320 ch, 6×12×6
L = 432
4 blocks
One Mamba block
at the bottleneck
input
layer norm
+x
−x
+y
−y
+z
−z
Mamba
sum
residual
output
each lane: 432 tokens
in its own order, its
own Mamba layer, then
unflatten
conv block
Mamba block
transposed conv
concatenate
Plates: footprint the grid, thickness the channels (log scales). L: sequence length per scan direction.